Practical Sampling

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Key Concepts

1) The conditions of the sampling theorem are rarely satisfied in practical applications.

    a) Many signals are not band limited.

    b) Noise is almost always present and is not band limited.

2) Practical sampling systems apply an analog or continuous-time low pass filter to the signal prior to sampling.

    a) The low pass filter is called an antialiasing filter.

b) The purpose of the antialiasing filter is to insure the signal is band limited and satisfies the conditions of the sampling theorem.

3) Practical antialiasing filters cannot transition from passband to stop band over a very small frequency range. That is, practical analog filters cannot have a brick wall frequency response.

a) The antialiasing filter passband is normally chosen to match the bandwidth $|\Omega| < W$ of the signal of interest.

b) The sampling theorem in satisfied by choosing the sampling frequency to exceed twice the frequency of the antialiasing filter stop band B, where $B > W$.

c) There is no aliasing into the signal band ($|\Omega| < W$) provided the sampling frequency exceeds $B + W$.

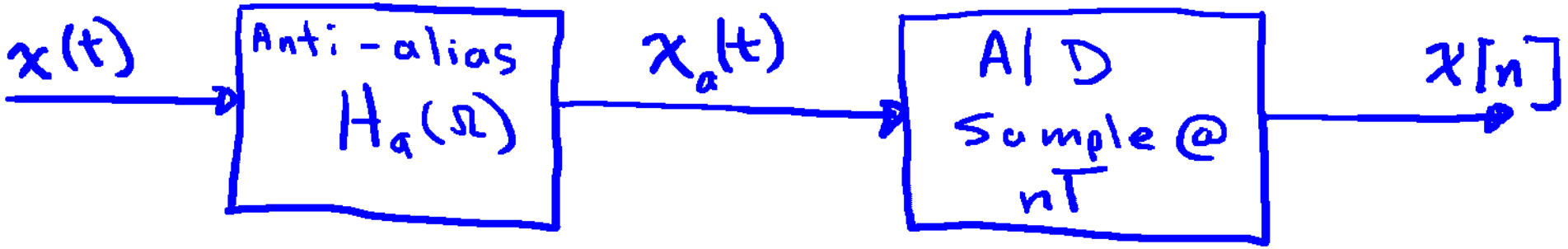
Video Screenshots

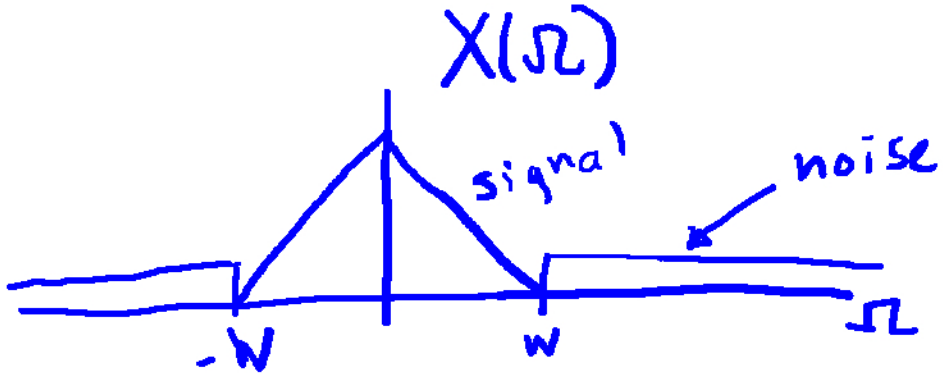
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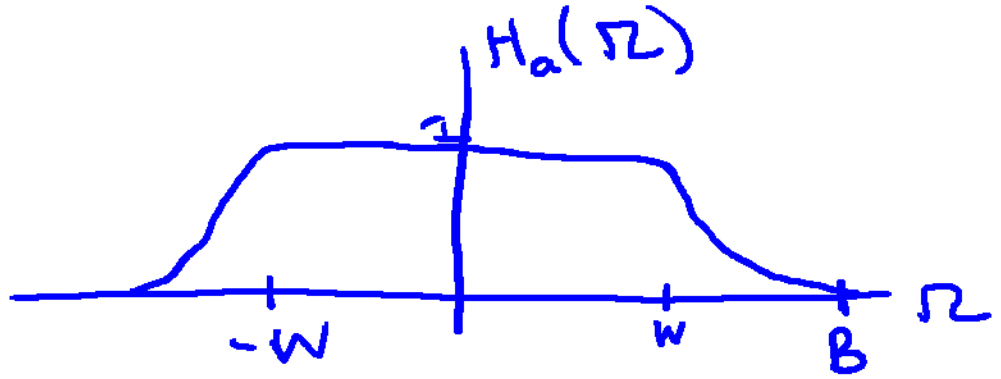
1) Many signals are not bandlimited

2) Noise is usually present

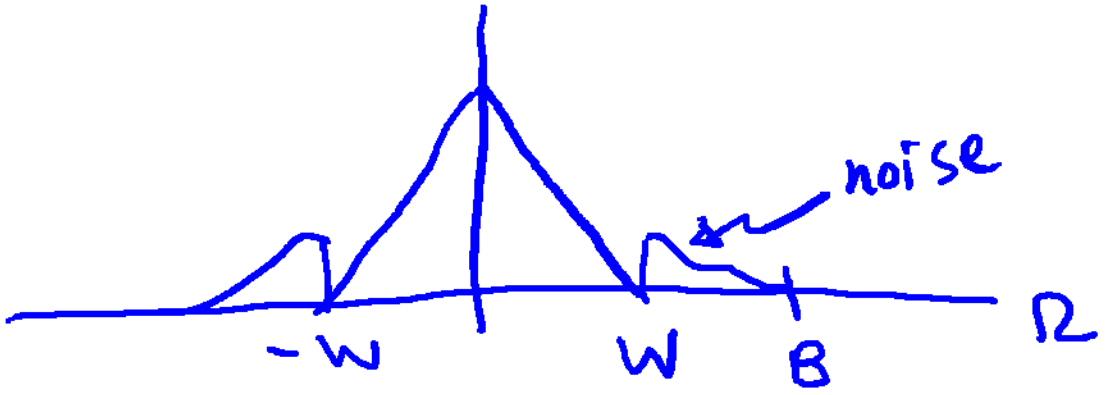
⇒ Use analog lowpass filter before sampling to guarantee the signal is bandlimited







$$X_a(\Omega) = X(\Omega)\, H_a(\Omega)$$



$\Omega_s = \frac{2\pi}{T} > 2B$

no aliasing

For no aliasing of noise into signal band

$\Omega_s - B > W \Rightarrow \Omega_s > B + W$

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